MUHAMMAD TALHA SHAHID

AI SOFTWARE ENGINEER

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TECHNICAL SKILLS

Languages	Python
LLMs & Agents	LangGraph, LangChain, Autogen, MCP Server, OpenAI API, Anthropic API, Groq, Multi-Agent Systems, Agentic Workflows, Tool/Function Calling, Prompt Engineering, Structured Outputs
RAG & Vector Search	RAG Pipelines, Text Embeddings, Semantic Search, ChromaDB, Pinecone, sqlite-vec
Evaluation & Testing	Benchmark Design, Baseline Comparison, Human-in-the-Loop Workflows, Claim Verification, Pydantic, pytest, Ruff, Mocked LLM Testing
Machine Learning	LSTM, Time-Series Forecasting, NLP, Sentiment Analysis, Computer Vision, Scikit Learn, Tensorflow
Backend & DevOps	FastAPI, Flask, REST APIs, Webhooks, SQLite, PostgreSQL, MongoDB, Firebase, Docker, Vercel, Git, GitHub, CI/CD

EDUCATION

BS Artificial Intelligence, Khwaja Fareed University of Engineering & IT, Rahim Yar Khan, Pakistan 2022 – 2026

- Participated in the 3-day International Conference on Emerging Trends in AI and Cyber Security (**ICEAC-2024**).

WORK EXPERIENCE

Machine Learning Intern, Ezitech Institute, Rawalpindi, Pakistan Jun 2025 – Sep 2025

- Developed an LSTM-based stock price prediction model on historical time-series data, achieving 88–90% accuracy.
- Implemented a multilingual NLP sentiment analysis system with 90%+ classification accuracy across languages.
- Built a computer vision smart traffic management system for vehicle detection with 90%+ detection accuracy.

PROJECTS

AI SRE Agent: Agentic Incident Response System

Python 3.12, *LangGraph*, *FastAPI*, *Flask*, *Pydantic*, *SQLite*, *httpx*, *GitHub REST API*, *GitHub Actions*, *OpenAI API*, *Anthropic API*, *pytest*, *Ruff*

- Designed a LangGraph incident-response agent that analyzes errors, finds root causes, and proposes code patches.
- Engineered human-in-the-loop repairs with SHA-256 patch approval, allow-lists, protected paths, and audit logs.
- Automated repair branches, Ruff and pytest runs via GitHub Actions, and draft PRs after explicit human approval.
- Delivered a FastAPI and Flask app with a live nine-step workflow view, diff viewer, and HMAC-verified webhooks.
- Verified with 42 tests and a 7-case benchmark, solving 6 of 7 with zero safety failures vs. 1 of 7 for no-AI baseline.

JobPilot: MCP-Powered Job Search Agent

Python, *FastMCP*, *OpenAI API*, *SQLite*, *sqlite-vec*, *Pydantic*, *Streamlit*, *Docker*

- Developed a multi-tool AI agent with six custom MCP servers for job discovery, matching, CV tailoring and tracking.
- Established an agent harness with tiered permissions, human approval, prompt-injection defenses, and cost limits.
- Managed context via token-budgeted assembly, rolling-summary compaction, and semantic and episodic memory.
- Added a claim-verification layer that checks generated CV bullets against profile facts to cut unsupported claims.
- Built an evaluation framework comparing LLM job matching to a TF-IDF baseline using labeled data and ablations.

EngiGraph: Multi-Agent RAG Assistant for Equipment Manuals

Python, LangGraph, LangChain, Groq API, Google Embeddings, ChromaDB, PyMuPDF, Pydantic, Flask, pytest

- Built a LangGraph investigator, planner, and verifier workflow for diagnostic reports with page-level citations.
- Constructed a RAG pipeline with page-aware chunking, Google embeddings, ChromaDB, and resumable indexing.
- Introduced structured-output validation, malformed-JSON retries, citation checks, and safety/uncertainty notes.
- Shipped a Flask app with PDF upload, background jobs, live progress, per-claim citations, and Markdown export.
- Authored a 57-question verified benchmark comparing accuracy, citations, latency, and cost to a RAG baseline.
- Wrote 30+ automated tests with mocked LLM and retrieval components, enabling offline testing at zero API cost.

FINAL YEAR PROJECT

ShopGen AI: WhatsApp-Integrated Conversational Shopping Agent 2025 – 2026

Python, LangGraph, LangChain, OpenAI GPT-4o Mini, Whisper, Whapi, Pinecone, Firebase, FastAPI, SerpAPI

- Architected a WhatsApp multi-agent system in LangGraph for product discovery, price comparison, and styling.
- Implemented stateful agent memory for multi-turn chats and preference tracking across 100+ interactions.
- Integrated Pinecone semantic product search and Firebase Realtime Database for inventory and sessions.
- Enabled text and voice chat (Whisper speech-to-text, text-to-speech) via Whapi with under 3-second latency.

CERTIFICATIONS

- Career Essentials in Generative AI, Microsoft & LinkedIn Learning
- AI Agents with MongoDB, MongoDB